

Fall-Detection Engineering Decision Matrix

Model	Accuracy 35%	Recall 25%	Runtime Efficiency 15%	Mobile Deploy. 15%	Interp. 10%	Weighted Score
Gradient Boosting	4.96 (99.18%)	5.00 (100.00%)	4.00 (100 stages)	4.00 (compact ensemble)	4.00 (staged trees)	4.586
SVM (RBF)	4.89 (97.90%)	4.92 (98.32%)	4.50 (1 kernel eval)	4.00 (JSON export)	3.00 (RBF margin)	4.517
Random Forest	4.95 (99.07%)	4.99 (99.76%)	3.00 (300 trees)	3.00 (large ensemble)	4.00 (feature import.)	4.281
Extra Trees	4.96 (99.18%)	4.99 (99.76%)	3.00 (300 trees)	3.00 (large ensemble)	3.50 (feature import.)	4.233

Scores are normalized on a 1-5 scale; bracketed values show the measured value or scoring basis.